Technical confidence is critical for secure identity systems, but cannot fully replace human governance. Even the most secure authentication system does not guarantee that a policy is correct or fair. While we should strive to provide confidence in as many situations as possible, doing so to the exclusion of trust isn't possible. The takeaway for dynamic authorization is that we should combine technical confidence with policy governance to create both secure and effective access control.

2.7 Privacy, authenticity, and confidentiality

Authorization systems play a big role in protecting privacy, ensuring authenticity, and providing confidentiality. These three concepts are interrelated, and understanding them precisely and the tradeoffs between them can help us design authorization systems and craft policies to correctly authorize access.

2.7.1 Distinguishing privacy from confidentiality

When people talk about privacy informally, they typically mean both privacy and confidentiality. But we will distinguish between these concepts so that we're clear about which problems we're solving and what tools to use. The International Association of Privacy Professionals (IAPP) glossary (https://iapp.org/resources/glossary/) defines four classes of privacy. Two of them, communications privacy and information privacy, are the most relevant to identity systems. *Communications privacy* is concerned with protecting the means of correspondence. *Information privacy* is concerned with the ability of individuals to consent to when, how, and to what extent information about them is communicated to others.

Another way of saying "the means of correspondence" is metadata—all the information in a communication that makes it possible. So, communications privacy is a metadata problem. Confidentiality is concerned with the content. Authenticity is knowing with confidence who the parties to a conversation are—and being able to prove that to others.

To understand these distinctions better, consider Alice, who visits a virtual health clinic at `https://ourvirtualhealth.com` for an anonymous